DATA ANALYTICS AND VISUALIZATION (24DSB3101)

Machine Learning-Based Healthcare Data Analytics for Early Disease Risk Predication

1. Problem Statement

Healthcare datasets contain demographic, clinical, and lifestyle attributes that can help identify individuals at elevated risk of disease. However, missing values, inconsistent records, class imbalance, and complex relationships among health factors can make reliable prediction challenging. Recent research also identifies concerns around model interpretability, dataset diversity, clinical workflow integration, and real-world validation. This project proposes an end-to-end healthcare analytics framework that preprocesses benchmark health datasets, develops disease-risk prediction models, evaluates them using standard metrics, and presents risk assessment through an interactive dashboard.

2. Literature Survey

Recent journal and review papers were selected to understand predictive healthcare, explainable AI, disease-risk modelling, and clinical decision-support research. The survey prioritizes 2024–2025 literature and framework/systematic-review studies.

No.	Research Paper	Year	Main Focus	Limitations
1	The role of explainable artificial intelligence in disease prediction: A systematic literature review and future research directions	2025	Reviews XAI techniques used for disease prediction and identifies research challenges.	Limited dataset diversity, model complexity, and dependence on single data types. (PubMed)
2	The application of explainable artificial intelligence in the prediction, diagnoses, treatment, and	2025	Studies XAI applications across chronic diseases; SHAP and LIME are widely used.	Limited multimodal data, limited data volume, and insufficient real-world clinical validation.

	management of chronic diseases: A systematic review			(PubMed Central (PMC))
3	Presenting Artificial Intelligence Predictions Based on Electronic Medical Records to Clinicians in Hospitals: A Systematic Review	2025	Reviews how AI predictions from structured healthcare records are presented to clinicians.	Limited evidence on optimal presentation methods and practical clinical integration. (PubMed)
4	How Explainable Artificial Intelligence Can Increase or Decrease Clinicians' Trust in AI Applications in Health Care: Systematic Review	2024	Examines the effect of XAI explanations on clinician trust in healthcare AI.	Explanations can sometimes increase excessive trust or decrease trust if they are unclear or oversimplified. (DOI)
5	Towards an Evaluation Framework for Explainable Artificial Intelligence Systems for Health and Well-being	2025	Proposes an evaluation framework for explainable AI systems in health and well-being.	Evaluation of XAI systems remains challenging, particularly for transparency and human understanding of AI decisions. (arxiv.org)

3. Research Gap Identification

This is very important for your project because your faculty specifically asks what previous research has not adequately addressed.

- Gap 1 — Disease-specific approaches
  Many existing studies concentrate on one disease or one dataset, making it difficult to establish a reusable healthcare analytics workflow.
- Gap 2 — Limited data diversity
  Recent literature identifies limited dataset diversity and reliance on single data types as important limitations in disease-prediction research.
- Gap 3 — Prediction vs. interpretability
  Many systems prioritize predictive performance, while explainability, transparency and trust are treated as secondary components.

- Gap 4 — Lack of integrated XAI workflow
  SHAP/LIME are frequently applied individually, but there is scope for a standardized workflow combining prediction, global explanation, local explanation and visualization.
- Gap 5 — Limited practical presentation
  Research often reports model performance without providing an accessible interface through which users can inspect risk predictions and their contributing factors. The literature on presenting AI predictions to clinicians highlights this practical issue.

4. Objectives

- To analyze healthcare datasets using Pandas, NumPy, Matplotlib, Seaborn, and automated profiling techniques to understand data distributions, correlations, data-quality issues, and significant disease-risk factors.
- To develop disease-risk prediction models using Logistic Regression, Decision Tree, Random Forest and XGBoost, and evaluate them using Accuracy, Precision, Recall, F1-Score and ROC-AUC.
- To improve the interpretability of disease-risk predictions using SHAP-based global and local feature explanations, enabling identification of the major factors influencing individual predictions.
- To develop an interactive Streamlit-based healthcare analytics dashboard that presents disease-risk probabilities, important risk factors, model performance and explainable prediction results for decision support.
- To evaluate the proposed healthcare data analytics framework comprehensively based on data quality, prediction performance, model interpretability, and overall system effectiveness.

5. Innovation

- End-to-End Healthcare Analytics – Integrates data preprocessing, analysis, prediction, and visualization.
- Multi-Model Prediction – Compares multiple ML models with XGBoost for disease-risk prediction.
- Explainable AI – Uses SHAP to identify the factors influencing each prediction.
- Interactive Dashboard – Uses Streamlit to provide an easy-to-use healthcare prediction interface.
- Visual Risk Insights – Presents risk probability, important features, and analytical charts clearly.

6. Novality

- Unified Healthcare Analytics Framework – Integrates data preprocessing, EDA, feature engineering, ML prediction, explainability, and visualization in one workflow.

- Multi-Disease Risk Prediction – Applies the framework to multiple healthcare datasets such as heart disease, diabetes, and stroke.
- Explainable Risk Prediction – Uses SHAP to identify the major health factors contributing to each prediction.
- Risk-to-Insight Analysis – Provides both the disease-risk probability and the key factors influencing the predicted risk.
- Interactive Decision-Support Dashboard – Uses Streamlit to present predictions, visual analytics, model performance, and explanations in one interface.
- Systematic Model Evaluation – Evaluates multiple ML models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC to select the most suitable model.

7. Feasibility

Technical Feasibility

The project can be implemented using publicly available healthcare datasets and Python-based open-source libraries such as Pandas, Scikit-learn, XGBoost, SHAP and Streamlit.

Data Feasibility

Public benchmark datasets are available for heart disease, diabetes and stroke prediction, making initial experimentation possible without requiring private hospital records.

Implementation Feasibility

The system can initially be developed using Jupyter Notebook/VS Code and subsequently integrated into a Streamlit dashboard.

Phase 1 → Literature Survey & Dataset Collection

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Phase 2 → Data Cleaning & EDA

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Phase 3 → Feature Engineering

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Phase 4 → ML Model Development

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Phase 5 → Model Evaluation

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Phase 6 → SHAP Explainability

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Phase 7 → Streamlit Dashboard

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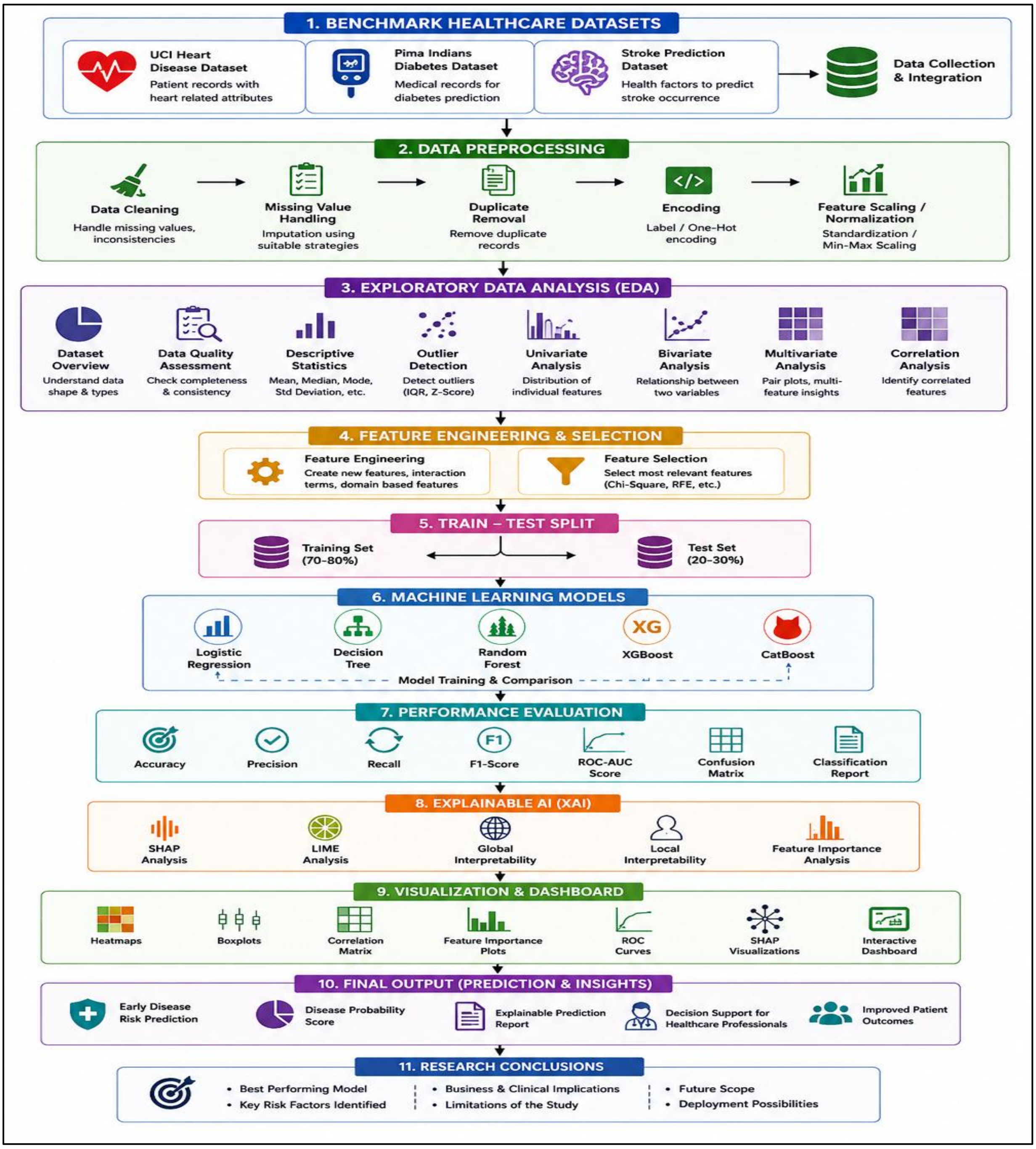
Phase 8 → Testing, Documentation & Final Report

8. Project Plan

Phase	Phase Name	Description
P1	Literature Survey & Problem Analysis	Study recent healthcare ML and XAI research papers, understand existing approaches, identify limitations, and finalize the research gap and problem statement.
P2	Dataset Collection & Data Understanding	Collect benchmark healthcare datasets such as UCI Heart Disease, Diabetes, and Stroke datasets and analyze their attributes, target variables, data types, and quality.
P3	Data Preprocessing & Exploratory Data Analysis	Handle missing values, duplicates and outliers, perform encoding/scaling where required, and analyze distributions, correlations and important health-risk patterns using EDA techniques.
P4	Feature Engineering & ML Model Development	Select or create relevant features and develop disease-risk prediction models using Scikit-learn algorithms and XGBoost.
P5	Model Evaluation & Explainability	Evaluate models using Accuracy, Precision, Recall, F1-Score and ROC-AUC, select suitable models, and use SHAP to explain important factors behind predictions.
P6	Dashboard Development & System Integration	Integrate the selected ML model and SHAP explanations into a Streamlit dashboard to display disease-risk probability, contributing factors and analytical visualizations.

9. Architecture

- **Benchmark Healthcare Datasets**
  Collect healthcare datasets such as UCI Heart Disease, Diabetes, and Stroke datasets for model development.
- **Data Preprocessing**
  Clean the collected data by handling missing values, duplicates, outliers, categorical values, and scaling where required.
- **Exploratory Data Analysis (EDA)**
  Analyze the datasets using statistics, distributions, correlations, and visualizations to identify important health-risk patterns.
- **Feature Engineering & Selection**
  Create or transform useful features and select the most relevant health attributes for prediction.
- **Train-Test Split**
  Divide the processed dataset into training and testing data to train and evaluate the models fairly.
- **Machine Learning Models**
  Develop prediction models using Logistic Regression, Decision Tree, Random Forest, and XGBoost.
- **Performance Evaluation**
  Compare the models using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix.
- **Explainable AI (XAI)**
  Use SHAP to explain which health factors contribute most to the model's risk prediction.
- **Visualization & Dashboard**
  Use Matplotlib, Seaborn, and Streamlit to present predictions, risk factors, model results, and visual insights.
- **Final Output / Decision Support**
  Provide an early disease-risk prediction, probability score, and important contributing factors through the dashboard.



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